

Top-Level Workplan – BFrep Manuscript Revision (Post-Pilot)

July 2026

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1.1 Overview

This document coordinates three parallel work streams for salvaging the manuscript “A pluralist perspective on input-output modeling: searching for commensurability” (Kapeller & Scharnreitner, submitted to *Metroeconomica*).

The revision is based on the finding that the existing code base ((3)BeyondHulten) is approximately 70–80% functional – far more advanced than earlier assessments assumed. This reduces the total timeline from 12–16 weeks to 6–10 weeks.

Update (2026-07-29): The four critical coding tasks from the Priority Matrix are **complete and verified** on Julia 1.12.6. The Go/No-Go decision is **GO**. The critical path has shifted from model implementation to manuscript writing and integration. Remaining timeline is estimated at 4–6 weeks.

1.2 Completed Milestones (This Round)

1.2.1 Mobile Labor + Economy-Wide Wage

Replaced sector-specific (immobile) wages with a single economy-wide wage $\mathbf{w}$. Sectoral labor allocation is now endogenous via the marginal product condition. Unknown vector grew from $2N$ to $2N+1$, with a numeraire constraint ($\text{CPI} = 1$) replacing one price equation to break the price-level indeterminacy inherent in CRTS models.

Key structural insight: Under CRTS + mobile labor, the zero-profit condition determines $(\mathbf{p}, \mathbf{w})$ as a function of technology parameters alone. The labor supply elasticity only affects the **scale** of the economy (total employment, GDP, sectoral quantities) – **not** relative prices or the wage. Verified empirically: price CV = 0.0000 across the full sweep, while quantity CV > 0.01 in 26/71 sectors. This is a standard CGE result and an important finding for the manuscript: the “labor supply elasticity continuum” is about aggregate effects, not sectoral price reallocation.

1.2.2 Parameter (Labor Supply Elasticity)

Implemented $\mathbf{L} = \mathbf{L} \cdot \hat{\mathbf{w}}$ as the labor supply function. The labor market clearing equation $\Sigma \mathbf{L}_i = \mathbf{L} \cdot \hat{\mathbf{w}}$ pins down the economy’s scale.

- $= 0$: perfectly inelastic (fixed total supply, Leontief-like)
- $\rightarrow \infty$: perfectly elastic (full labor slack)
- $0 < < \infty$: intermediate continuum

1.2.3 Variance Decomposition

Full factorial design over $\times \times \times$ ($5 \times 3 \times 3 \times 3 = 135$ evaluations) with ANOVA-style partial R^2 :

Factor	Partial R^2	Share
	0.859	88.4%
	0.075	7.7%
	0.034	3.5%
	0.004	0.4%
Total	0.971	
Residual (interactions)	0.029	

dominates – it accounts for ~88% of explained variance in real GDP.

1.2.4 Pilot Sweep — GO

Criterion	Result	Threshold	Pass?
Sectors with CV > 0.01	26/71	5	
Max sectoral CV	0.057	>0.01	
partial R ²	0.859	>0.05	

Top-5 most affected sectors (by quantity CV): 1. Grundstücks- u. Wohnungswesens (CV=0.057) 2. Kraftwagen und Kraftwagenteile (CV=0.049) 3. Vorb. Baustellen-, Bauinstallations-, Ausbauarbeiten (CV=0.040) — the shocked sector 4. Großhandelsleistungen (CV=0.034) 5. Öffentliche Verwaltung (CV=0.031)

1.2.5 Julia Environment Aligned to Host (1.12.6)

- **Project.toml**: added [compat] julia = "~1.12"
- **Manifest.toml**: targets julia_version = "1.12.6"
- Julia 1.12.6 Linux ARM64 binary persists at /root/.julia/linux-julia-1.12.6/ (mounted host filesystem)
- Minimal test project (**tests/minimal_test/**) runs all tests successfully
- Code is now portable between container (Linux) and host (macOS)

1.2.6 Bug Fix

Fixed Julia destructuring bug in `mobile_labor.jl`: `(; supply_shock, factor_share) = (shocks, data)` was creating a plain tuple instead of accessing struct fields. Corrected to separate `(; supply_shock) = shocks` and `(; factor_share) = data`.

1.3 Work Streams

1.3.1 Stream A: Manuscript Revision

Location: (1)Submission/revise/

Status: Scaffold created. The manuscript has been reorganized into a modular structure:

```
revise/
---- main.tex                -- new main file with revised abstract
---- BandF.bib               -- bibliography (copied)
---- pictures/               -- figures (copied)
---- chapters/
---- ch01_introduction.tex   -- % OLD TEXT FROM HERE ON
---- ch02_io_leontief.tex    -- % OLD TEXT FROM HERE ON
---- ch03_cge_model.tex      -- % OLD TEXT FROM HERE ON
---- ch04_data.tex           -- % OLD TEXT FROM HERE ON
---- ch05_ces_functions.tex  -- % OLD TEXT FROM HERE ON
---- ch06_labor_slack.tex    -- % OLD TEXT FROM HERE ON
---- ch07_conclusion.tex     -- % OLD TEXT FROM HERE ON
---- ch08_backmatter.tex     -- % OLD TEXT FROM HERE ON
```

The new abstract replaces the “paradigmatic cleavage” framing with a “labor supply elasticity continuum” framing, centered on Robinson (2006) and the quantitative decomposition of closure choice vs. technology choice.

Remaining work:

- Rewrite introduction (drop Kuhnian preamble to one paragraph, open with policy question, state upfront that IO is a special case of CGE)
- Rewrite Section 2 (streamline IO exposition, fix sign error in eq. 19, remove “supply-side vs demand-side” distinction, be transparent about open economy)
- Rewrite Section 4 (drop CES novelty claim, drop Cobb-Douglas as “archetypical neoclassical”)

- **Replace Section 5** (exogenous labor slack → endogenous labor supply with η parameter) — **model code is ready, results verified**
- **Rewrite Section 6** (drop “deus ex machina” hand-wringing, present variance decomposition results: $\eta = 88.4\%$ of explained variance) — **results are ready**
- Integrate new literature: Robinson (2006), Rose (1995), Dervis et al. (1982), de Melo & Tarr (1992), McGregor et al. (1996), Shoven & Whalley (1984)
- Fix minor issues (p.5 sentence, p.16 paragraph, Figure 3 axis)

Dependencies: ~~Requires results from Stream B (η sweep, variance decomposition) before Sections 5–6 can be finalized.~~
RESOLVED: Stream B results (η sweep, variance decomposition, Go/No-Go) are complete. Sections 5–6 can proceed immediately.

1.3.2 Stream B: Model Extension (BeyondHulten code base)

Location: (3)BeyondHulten/src/

Status: Core code is 70–80% functional. Three model types (CES, Leontief, Cobb-Douglas) work. German data is connected. Labor slack is implemented as a function parameter. Elasticity gradient and plotting infrastructure exist.

Completed this round: - Implement continuous labor supply elasticity η in $[0, \infty)$ — done in `mobile_labor.jl` - Switch from sector-specific to mobile labor with economy-wide wage — done in `mobile_labor.jl` - Implement variance decomposition ($\eta \times \varepsilon \times \theta \times \sigma$ factorial with partial R^2) — done in `variance_decomposition.jl` - Go/no-go decision: pilot η sweep — **GO** ($\eta = 88.4\%$ of variance, 26/71 sectors with $CV > 0.01$)

Remaining work:

- Open the economy (imports, exports, domestic vs. imported intermediates) or document the closed-economy mapping transparently
- Fix calibration documentation (summary table of α , β , λ , Domar weights, elasticities)
- Update notebooks to match current API (some notebooks use older function signatures)
- Generate final figures for the manuscript (η sweep plots, variance decomposition bar chart, sectoral variation heatmap)
- ~~Full η sweep + variance decomposition~~ — **DONE**
- ~~Implement η parameter~~ — **DONE**
- ~~Switch to mobile labor~~ — **DONE**

Dependencies: ~~Can proceed immediately on existing code.~~ **Critical path cleared.** The B&F replication (Stream C) provides validation but is not a prerequisite.

1.3.3 Stream C: B&F Replication

Location: (3)BeyondHulten/bf_replication/

Status: Workplan created (REPLICATION_WORKPLAN.md). No code yet.

Purpose: A clean-room replication of Baqaee & Farhi (2019) on US data. Serves two functions:

- Validates the model implementation against published results
- Provides the computational infrastructure for the “replicate and contextualize” strategy from the salvage plan

Remaining work: See `bf_replication/REPLICATION_WORKPLAN.md` for the 6-phase plan (R1–R6).

Dependencies: Can reuse code patterns from `BeyondHulten/src/`. Independent of Streams A and B.

1.4 Cross-References

This workplan	References
Full salvage strategy	(4)plans/REVISED_SALVAGE_PLAN.md
Earlier plans (superseded)	(4)plans/SALVAGE_PLAN.md, salvage_plan2.md, salvage1_comment.md, BF_CONTRIBUTION_RECEPTION.md
Replication details	(3)BeyondHulten/bf_replication/REPLICATION_WORKPLAN.md
Manuscript source	(1)Submission/revised/main.tex + chapters/
Original manuscript	(1)Submission/foundations_of_pdf/submission(1).tex

This workplan	References
Reviews	(2)Reviews/metro-rev1.docx, metro-rev2.docx
Core code	(3)BeyondHulten/src/
Original workplan	(3)BeyondHulten/WORKPLAN.md
Model source files	src/mobile_labor.jl, src/variance_decomposition.jl
Test suite	tests/minimal_test/ (run_test.jl, test_full_pilot.jl)
Julia modeling skill	Hermes skill beyond-hulten-julia-modeling

1.5 Execution Order (Revised)

1.5.1 ~~Weeks 1–2: Foundation~~ — COMPLETED

- ~~Stream A:~~ Draft new introduction, rewrite Section 2 (pending)
- ~~Stream B:~~ Implement η parameter, switch to mobile labor, run pilot sweep — **DONE**
- **Stream C:** Phase R1 (data pipeline), start Phase R2 (core solver)

1.5.2 ~~End of Week 2: Go/No-Go~~ — DECISION: GO

- Check pilot η sweep results: are sectoral dynamics interesting? — **YES** (26/71 sectors, CV up to 0.057)
- Check variance decomposition: does η dominate $(\varepsilon, \theta, \sigma)$? — **YES** (= 88.4% of variance)
- If yes to either: proceed with full plan — **PROCEEDING**
- If no to both: pivot to pure replication strategy — **NOT NEEDED**

1.5.3 Weeks 1–2: Writing Sprint (Current Phase)

- **Stream A:** Draft new introduction, rewrite Section 2 (fix sign error, remove “supply vs demand”, state model closure clearly)
- **Stream A:** Write Section 5 (endogenous labor supply with) — results ready
- **Stream A:** Write Section 6 (variance decomposition results) — results ready
- **Stream B:** Generate final figures (sweep, variance decomposition, sectoral variation)
- **Stream B:** Calibration documentation
- **Stream C:** Phase R1–R2 (data pipeline, core solver)

1.5.4 Weeks 3–4: Integration

- **Stream A:** Rewrite Sections 4–5 (CES, labor supply), integrate literature
- **Stream B:** Open economy (or transparent mapping), update notebooks
- **Stream C:** Phase R2 (solver), Phase R3 (shock simulation)

1.5.5 Weeks 5–6: Finalization

- **Stream A:** Write results section (aggregate, sectoral, variance decomposition), discussion, conclusion
- **Stream A:** Copy-edit, fix minor issues, finalize manuscript
- **Stream A:** Write response letter to both reviewers
- **Stream C:** Phase R4 (elasticity gradient), Phase R5 (reallocation) — if time permits

1.5.6 Week 7–8: Submission

- Final review of all three deliverables
- Submit revised manuscript + response letter + replication code

1.6 Priority Matrix (Revised)

Task	Stream	Priority	Status	Effort
Implement η parameter	B	Critical	DONE	3–5 days
Mobile labor (economy-wide wage)	B	Critical	DONE	2–3 days
Variance decomposition	B	Critical	DONE	2–3 days
Pilot η sweep	B	Critical	GO	1 day
New introduction	A	High	Pending	2 days
Rewrite Section 2 (fixes)	A	High	Pending	2 days
Write Section 5 (labor supply with)	A	High	Pending — results ready	2–3 days
Write Section 6 (variance decomposition)	A	High	Pending — results ready	2 days
Literature integration	A	High	Pending	2–3 days
Generate final figures	B	High	Pending	1–2 days
Calibration documentation	B	Medium	Pending	1 day
Open economy	B	Medium	Can be transparent mapping	3–5 days
Rewrite Sections 4	A	High	Pending	2–3 days
Results section	A	High	Needs figures	2–3 days
Response letter	A	High	Needs final manuscript	2–3 days
Replication R1–R2	C	Medium	Validates model	5–10 days
Replication R3–R6	C	Low	Nice to have	10–15 days
Update notebooks	B	Low	Maintenance	1–2 days

1.7 Risk Register (Revised)

Risk	Likelihood	Impact	Mitigation	Status
η sweep results are smooth/monotonic	Medium	High	Size claim is insurance	RESOLVED: dominates (88.4%), 26/71 sectors show variation
η affects only scale, not prices	—	Medium	Embrace as finding: standard CGE result, strengthens “closure choice” framing	NEW — key insight for manuscript
Reviewer 2 irreconcilable	Medium	High (desk reject)	Maximally gracious response; size claim is genuine contribution	Unchanged

Risk	Likelihood	Impact	Mitigation	Status
Mobile labor implementation difficult	Low	Medium	Well-understood CGE modification	RESOLVED: implemented and verified
Open economy too hard	Medium	Medium	Fallback: transparent documentation of closed-economy mapping	Unchanged
Replication takes too long	Medium	Low	Replication is not blocking; model extension is complete	Unchanged
Full project (GLMakie etc.)	High	Low	Run on host Mac (more RAM); use minimal_test for container work	NEW
OOM in container				

1.8 Reviewer Response Strategy

1.8.1 Reviewer 2 (devastating, recommends rejection)

- Acknowledge core criticism fully: “the bridge already exists” (Robinson 2006)
- Show endogenous labor supply replaces exogenous injection — **now implemented and verified**
- Present variance decomposition as genuinely new quantitative finding — **= 88.4% of explained variance**
- Address every specific point (sign error, closed economy, immobile labor, CES novelty, etc.)
- Be maximally gracious – do not defend the original framing
- **New ammunition:** The CRTS result (affects scale, not prices) is a clean theoretical finding that demonstrates understanding of CGE closure properties

1.8.2 Reviewer 1 (moderate, constructive)

- Address each point systematically
- Show restructured paper with clear model description
- Include calibration summary table
- Engage with McGregor et al. (1996)
- Clarify aggregate vs sectoral distinction (now quantified via variance decomposition: = 88.4% aggregate, 26/71 sectors show sectoral variation)